

Jech 1

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1 Introduction

1.1. Suppose $\{\{a\}, \{a, b\}\} = \{\{a'\}, \{a', b'\}\}$. Then $\{a\} = \{a'\}$ (1) or $\{a\} = \{a', b'\}$ (2). If (1), then $a = a'$. $\{a, b\} = \{a'\}$ (1.1) or $\{a, b\} = \{a', b'\}$ (1.2).

If (1.1), $a = a'$ and $b = a'$, so $a = b$. Rewrite the LHS from Axiom of Extension as $\{\{a\}, \{a\}\} = \{\{a\}\}$. Then $\{a', b'\} = \{a\}$, so $a' = a$ and $b' = a$, so $a = b = a' = b'$.

If (1.2), $b = a'$ (1.2.1) or $b = b'$ (1.2.2). If (1.2.1), $a = a' = b$, so rewrite the LHS as $\{\{a\}\}$. Proceed with (1.1). Else in (1.2.2) $a = a'$ and $b = b'$.

If (2), then $a = a'$ and $a = b'$, so $a' = b'$. This case is analogous to (1.1).

These cases are sufficient since if $a = a'$ and $b = b'$, then $\{a\} = \{a'\}$ and $\{a, b\} = \{a', b'\}$, so $\{\{a\}, \{a, b\}\} = \{\{a'\}, \{a', b'\}\}$.

1.3: If X is inductive, then the set $S = \{x \in X : x \subset X\}$ is inductive.

$\emptyset \in X$ by definition of inductive and also $\emptyset \subset X$, so $\emptyset \in S$. Let $s \in S$. $s \cup \{s\} \in X$ as $s \in X$ and X is inductive. $s \in X$ and $s \subset X$ by construction of S , so $s \cup \{s\} \subset X$. Hence, $s \cup \{s\} \in S$, so S is inductive.

Define $N = \bigcap \{X : X \text{ is inductive}\}$. Then N is inductive. $\emptyset \in X$ for all inductive X , so $\emptyset \in N$. Let $u \in N$. For all inductive X , $u \cup \{u\} \in X$, so $u \cup \{u\} \in N$. N is inductive.

Let $u \in N$ and X be inductive. Each X has a corresponding inductive $S \subset X$ with the property $u \in S \rightarrow u \subset X$. By construction of N , $u \in S$ as well, so $u \subset X$ for all X , or that $u \subset N$. Hence, N is transitive.

Let $n \in N$ be arbitrary. Show that $n = \{m \in N : m \subset n\}$???

1.4: If X is inductive, then $S = \{x \in X : x \text{ is transitive}\}$ is inductive. $\emptyset \in X$ and $\emptyset$ is transitive, so $\emptyset \in S$. Suppose $s \in S$. Then $s \cup \{s\} \in X$ as X is inductive

and $S \subset X$. Suppose $a \in s \cup \{s\}$. If $a \in s$, then $a \subset s$ as s is transitive. It follows $a \subset s \cup \{s\}$. Else $a = s$, for which $a \subset s \cup \{s\}$ as well. Hence, $s \cup \{s\}$ is transitive, so $s \cup \{s\} \in S$ and S is inductive.

It follows every $n \in N$ is transitive.

1.5: Add the additional condition $x \not\subset x$. Let X be inductive and let $S = \{x \in X : x \text{ is transitive and } x \not\subset x\}$. Show that S is inductive.

$\emptyset \in X$, $\emptyset$ is transitive, and $\emptyset \not\subset \emptyset$. Suppose $u \in S$. Sps FTSOC $u \cup \{u\} \in u \cup \{u\}$. Then $u \cup \{u\} = u$, from which $u \in u$, contradiction, or $u \cup \{u\} \in u$, from which $u \in u$ as well, contradiction. Hence, $u \cup \{u\} \notin u \cup \{u\}$.

It follows $n \not\subset n$. If $n \cup \{n\} = n$, then $n \in n$ as well, contradiction. Hence, $n + 1 \neq n$.

1.6. Let X be inductive and let

$S = \{x \in X : x \text{ is transitive and for all nonempty } s \subset x \text{ there exists an } \in\text{-minimal element } t \in s\}$.

Show that S is inductive.

$\emptyset \in S$. Let $u \in S$ and consider $u \cup \{u\}$. Suppose $s \subset u \cup \{u\}$, so that $s \subset u$ or $s = t \cup \{u\}$, where $t \subset u$. The former case works. In the second case, if t is nonempty pick the $\in$ -minimal element of t , $t \subset u$. Else $s = \{u\}$, which has only one element and is $\in$ -minimal (assuming $u \not\subset u$). Hence, S is inductive.??????????

1.8. If X is inductive, then so is $S = \{x \in X : x = \emptyset \text{ or } x = y \cup \{y\} \text{ for some } y \in X\}$.

$\emptyset \in S$. Let $u \in S$. Then $u \cup \{u\} \in X$ as $u \in X$. Moreover by construction, $u \cup \{u\} \in S$. It follows for any $n \neq 0$, there exists $m \in N$ such that $n = m + 1$.

1.9. Let $A \subset N$ such that $0 \in A$ and if $n \in A$, then $n + 1 \in A$. Show $A = N$.

A by construction is inductive. By definition of N , $N \subset A$. By givens $A \subset N$. By axiom of Extensionality $A = N$.

1.10. Let $n \in N$. Then $\bigcup n \in N$ as well. If $n = \emptyset$, we're done. Else there exists m such that $n = m \cup \{m\}$. Note $n = \{u \in N : u \in n\}$. Since if $u \in n$, then $u \in m$ or $u = m$, we have $u \subset m$ by transitivity of N . It follows $\bigcup n = m \in N$.

Show for all $n \in N$, $n = \emptyset$ or $\emptyset \in n$. Let X be inductive. Define $S = \{x \in X :$

$x = \emptyset \vee \emptyset \in x$. Show that X is inductive.

$\emptyset \in S$. Suppose $u \in S$. Consider $u \cup \{u\}$. If $u = \emptyset$, we're done. Else, $\emptyset \in u \subset u \cup \{u\}$.

Let $S = \{x \in X : \forall y \in X (y \subset x \wedge y \neq x \rightarrow y \in x)\}$. Show that S is inductive. $\emptyset \in S$. Suppose $u \in S$. Let $t \subset u \cup \{u\}$. If $t = r \subset u$ and $r \neq u$, then $r \in u \subset u \cup \{u\}$. If $r = u$, then $u \in u \cup \{u\}$. If $t = r \cup \{u\}$ and $r \neq u$, then $r \in u$. $u \in u \cup \{u\}$, so $t \in u \cup \{u\}$?